Project Intermediate Report: ML Snake (Choose Your Own)

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Current state

- Playable Snake in pygame: menu, 15×15 grid, WASD controls, apples, growth, score, game over on wall or self collision.
- Board is a 2D list (snake cells hold age, apple = -1, empty = 0).
- A BOT state and numpy are in place; the ML agent is not built yet.

How I know it runs correctly

- Snake moves one cell per tick; 180° reversals are blocked.
- Wall and body collisions end the game; moving into the tail cell is allowed.
- Eating an apple grows the snake, adds 1 to the score, and spawns a new apple on an empty cell.
- Restart resets all state.

Remaining

- Separate game logic from rendering and add a headless mode.
- Build the agent (state, actions, reward, training loop).
- Evaluate against a random agent over many games.
- Fix known issues: spawnApple crash on a full board, leftover debug prints, empty README.